

Tarek Zaki

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Education

Alexandria University

B.Sc in Electrical Engineering

Sep 2017 – Jul 2022

Grade: Very Good

AWS re/Start Graduate, AWS

Cloud fundamentals and hands-on AWS training

May 2024 – Aug 2024

(Verified Credential)

Experience

Vodafone, Cairo

DevOps Engineer

May 2025 – Present

- Designing and automating scalable infrastructure, CI/CD pipelines, and internal platform services for high-availability enterprise environments.
- Supporting DevOps and data platform operations using Kafka, NiFi, Hadoop, and HDFS, including secure Kafka administration with SASL authentication and ACL-based authorization.
- Integrated big data and streaming technologies including Hive, Spark, Trino, Sqoop, Flink, and Kafka for analytics and real-time processing workloads.
- Built centralized observability and logging solutions using Prometheus, InfluxDB, Grafana, Logstash, and Elasticsearch for infrastructure monitoring, alerting, and system activity tracking.
- Working on Kubernetes-based platform engineering using kubectl, Rancher, ingress controllers, and monitoring stacks for internal platform and MLOps workloads.
- Contributed to internal MLOps platform setup using Kubeflow, ZenML, KServe, and Qdrant for model deployment and inference validation.
- Implemented workflow automation and self-service operational processes using n8n and Ansible to streamline deployments, notifications, and service provisioning.
- Working on IaC, compliance enforcement, platform reliability, and operational automation across hybrid infrastructure environments.

Ortoscale, Slovenia

DevOps & Observability Engineer, Freelance/Contract

March 2025 – Feb 2026

- Built Grafana dashboards with PromQL and VictoriaMetrics; integrated OpenSearch for log analytics.
- Automated observability stack provisioning and alerting for multi-tenant environments.
- Migrated microservices from Docker Swarm to Kubernetes using Helm and custom manifests.

Giza Systems, Cairo

Site Reliability Engineer

Sep 2024 – Feb 2025

- Built and managed Kubernetes clusters (multi-master, kubectl) for staging and production workloads.
- Developed a scalable messaging pipeline using MQTT, Kafka, and Elasticsearch.
- Implemented centralized logging with Fluent Bit, Loki, and Grafana.
- Automated CI/CD workflows with ArgoCD and Helm; explored OpenShift pipelines.
- Backed up clusters using Velero with MinIO while preparing for containerd upgrade, exploring update strategies.

Projects

Research-to-Production AI Platform

(GitHub)

- Built an end-to-end AI infrastructure lab spanning reproducible PyTorch training, model governance, HPC scheduling, Kubernetes serving, and observability.
- Containerized CUDA-enabled LSTM training; versioned datasets with DVC/MinIO and tracked code, data, configuration, runtime, metrics, and artifacts with MLflow.
- Benchmarked GPU training across batch sizes, mixed precision, CUDA memory behavior, data transfers, and CPU/GPU throughput.
- Built a three-node Slurm cluster demonstrating multi-node jobs, resource scheduling, contention, pending states, backfill, cancellation, and troubleshooting.
- Deployed PyTorch inference on a two-node K3s cluster with KServe, Knative, and Istio, validating scale-to-zero and cold starts.
- Implemented baseline-based model quality gates and Prometheus/Grafana monitoring for serving replicas, CPU, memory, and pod lifecycle.

Telecom Churn MLOps GitOps Platform

(GitHub)

- Built an end-to-end MLOps inference platform for a telecom churn prediction model using FastAPI, Docker, GitHub Actions, Docker Hub, Kubernetes, ArgoCD, and KServe.
- Automated model training, Docker image build, and registry publishing through GitHub Actions, with the image deployed from Docker Hub into a local kind Kubernetes cluster.
- Implemented GitOps-based deployment using ArgoCD and Kustomize to continuously reconcile Kubernetes manifests for the FastAPI inference service.
- Deployed the model through both a standard Kubernetes NodePort service and a KServe custom predictor, validating real-time inference through live API requests.

Serverless Event-Driven Image Processing Pipeline on AWS

(GitHub)

- Built an event-driven image processing pipeline using S3, EventBridge, SQS, Lambda, DynamoDB, and CloudFront.
- Implemented asynchronous processing with SQS, dead-letter queues, CloudWatch alarms, and X-Ray tracing for resilient operations.
- Deployed infrastructure using AWS SAM and CloudFormation with IAM least-privilege access and secure S3 delivery through CloudFront OAC.

DevOps CI/CD Pipeline with Infrastructure Hardening

(GitHub)

- Provisioned and configured three CentOS VMs with Vagrant for Jenkins, Gogs, and deployment infrastructure.
- Built a complete CI/CD pipeline: Gogs webhook → Jenkins pipeline → Ansible-based Apache deployment → Docker image export.
- Developed modular Ansible roles for secure Apache setup, log rotation, header hardening, and IP-based access control.
- Automated user creation, audit logging, service monitoring, and disk alerts using Bash, ‘auditd’, and hourly cron jobs.

Scalable Web Application on AWS

(GitHub)

- Deployed a highly available architecture with EC2 Auto Scaling, ALB, S3, and Route 53 across multiple AZs.
- Initially used self-managed MySQL, Memcached, and RabbitMQ; re-architected with RDS, ElastiCache, and Amazon MQ for scalability and maintainability.
- Automated deployment via Elastic Beanstalk with ACM, Route 53, CloudFront, and CloudWatch integration.

Skills

Languages: Bash, Python, Golang, C/C++
Operating Systems: Linux, Windows
Cloud Platforms: AWS, GCP, Azure
Containers & Orchestration: Docker, Kubernetes, OpenShift, Rancher
CI/CD & Automation: ArgoCD, Helm, Jenkins, Tekton, n8n, Ansible, Terraform
Monitoring & Logging: Datadog, ELK, Fluent Bit, Logstash, Grafana, Loki, Prometheus, InfluxDB
Messaging & Streaming: Kafka, Flink, Conduktor, Mosquitto, RabbitMQ
Big Data & Analytics: Hadoop, HDFS, Apache NiFi, Hive, Spark, Trino, Sqoop, ClickHouse
MLOps & AI: PyTorch, MLflow, DVC, MinIO, Kubeflow, ZenML, KServe, LiteLLM, Qdrant
HPC & GPU Infrastructure: Slurm, CUDA, NVIDIA GPU Monitoring, GPU Resource Scheduling
Version Control: Git
Spoken Languages: Arabic (Native), English (C1), German (B1), French (A1)

Certifications

Certified Kubernetes Administrator (CKA) <i>Issued by Linux Foundation</i>	2026 ☐ Verified Credential
AWS Certified Solutions Architect – Associate (SAA-C03) <i>Issued by Amazon Web Services</i>	2026 ☐ Verified Credential
Google Cloud Certified Professional Cloud Architect <i>Issued by Google Cloud</i>	2026 ☐ Verified Credential
Google Cloud Generative AI Leader <i>Issued by Google Cloud</i>	2026 ☐ Verified Credential
Google Cloud Certified Associate Cloud Engineer <i>Issued by Google Cloud</i>	2025 ☐ Verified Credential
Kubernetes and Cloud Native Associate (KCNA) <i>Issued by Linux Foundation</i>	2025 ☐ Verified Credential
AWS Certified Cloud Practitioner <i>Issued by Amazon Web Services</i>	2024 ☐ Verified Credential